Jobs Phase 2 — Services

Services that will be created or updated for Phase 2 (SRP, Detail Page, MyAccount).

Architecture Overview

```
Frontend (Next.js) / App (React Native)
    ↓ GraphQL queries/mutations
marketplace-graphql (GraphQL Gateway)
    ↓ REST API calls
marketplace-backend/apps/listing/services/listing-http-rest (Go REST API)
    ↓ Data operations
MongoDB → Mongo Connector → Elasticsearch
                            Redis (cache)
```

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1. listing-http-rest (Go REST API)

Location: `marketplace-backend/apps/listing/services/listing-http-rest`

Updates Required

1.1 Listing Model / Data Layer

- Add `deletedAt` (timestamp) field to the listing model
- Add 'deleted' as a valid listing status
- Add "Job" as a valid `marketType` value
- Verify `jobsPayRangeType`, `jobsPayFrom`, `jobsPayTo`, `jobsApplicationUrl` fields exist on the model (added in Phase 1 — may need expansion for reads)

1.2 Listing GET Endpoint (Detail Page)

- **Endpoint**: `GET /listings/{id}`
- **Change**: Expand response to include Jobs-specific top-level fields:
  - `jobsPayRangeType`
  - `jobsPayFrom`
  - `jobsPayTo`
  - `jobsApplicationUrl`
- specSubCat fields (employment type, education, experience, perks) should already be part of the listing response as specs

1.3 Search Service (SRP)

- **Endpoint**: Search endpoint in the dedicated SRP search service (already extracted from GraphQL)
- **Changes**:
  - Support `marketType=Job` filtering
  - Add custom filter mapping for pay range fields so `jobsPayFrom` / `jobsPayTo` can be used as range filters in SRP queries
  - Ensure specSubCat-based filters are included in AvailableFilters when Jobs category/subcategory is selected

1.4 New MyAccount Action Endpoints

These endpoints need to be created or verified on the Listing Service:

Endpoint	Method	Description
`/listings/{id}/mark-sold`	`PUT`	Mark listing as sold
`/listings/{id}/mark-sale-pending`	`PUT`	Mark listing as sale pending
`/listings/{id}/delete`	`DELETE`	Soft delete — sets status='deleted' + populates `deletedAt`
`/listings/{id}/renew`	`PUT`	Renew/extend listing expiration
`/listings/{id}/request-activation`	`PUT`	**Already exists** — reactivate a listing
`/listings/{id}/restore`	`PUT`	CX-only restore of soft-deleted listing (contract TBD pending CX input)

1.5 Soft Delete Behavior

When a listing is soft-deleted via the delete endpoint:

1. Set `status` to 'deleted'
2. Populate `deletedAt` with current timestamp
3. Remove from ES search index (or filter out via status)
4. Do not count toward user's active listing limits
5. Stop billing/subscriptions for the listing
6. Retain photos and all listing data for potential restore

When restored (by CX or user reactivation):

- CX restore: direct status change back, photos intact
- User reactivation: must go through checkout flow to re-pay for subscriptions, must be within listing limits

Retention period before final hard purge: **pending Legal input**

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2. marketplace-graphql (GraphQL Gateway)

Location: `marketplace-graphql`

Updates Required

2.1 Feature Flag — Jobs Category

- Add feature flag that filters Jobs category from production responses
- Allowlist based on member ID — only team members see Jobs category
- Flag must be removable for launch

2.2 SRP Query Updates

- Add "Job" to the `marketType` filter options returned to frontend

- Pass through Jobs specSubCat filters via existing AvailableFilters pipeline
- Add custom handling for pay range filter (top-level fields, not specSubCat)
- Proxy all filter/search requests to the dedicated SRP search service

2.3 Detail Page Query Updates

- Update listing detail query to pass through Jobs-specific top-level fields from the Listing Service:
  - `jobsPayRangeType`, `jobsPayFrom`, `jobsPayTo`, `jobsApplicationUrl`

2.4 MyAccount Mutation Updates

- Add/verify GraphQL mutations that proxy to the new Listing Service endpoints:
  - `markListingAsSold`
  - `markListingAsSalePending`
  - `deleteListing` (soft delete)
  - `renewListing`
  - `restoreListing` (CX-only — contract TBD)

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3. Elasticsearch

Updates Required

3.1 Index Mapping Changes

Add mappings for top-level Jobs fields:

Field	ES Type	Notes
`jobsPayRangeType`	`keyword`	"hourly" or "salary"
`jobsPayFrom`	`integer`	Stored in cents
`jobsPayTo`	`integer`	Stored in cents
`jobsApplicationUrl`	`keyword`	Not used for filtering — included for detail reads

Add "Job" as a valid value for the existing `marketType` field.

3.2 Soft Delete Exclusion

Ensure soft-deleted listings (status='deleted') are excluded from all search results. Either:

- Remove from ES index on soft delete, re-index on restore
- Or filter by status in all queries (existing pattern TBD)

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4. MongoDB / Mongo Connector

Updates Required

4.1 Listing Document Changes

- Add `deletedAt` field (nullable timestamp) to listing documents
- Verify 'deleted' is a valid status value in the schema

4.2 Mongo Connector Sync

- Update connector to sync new top-level fields to ES: `jobsPayRangeType`, `jobsPayFrom`, `jobsPayTo`, `jobsApplicationUrl`
- Verify specSubCat fields already sync via existing pipeline

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5. Category Manager (Admin)

Updates Required

- Create Jobs category
- Create Jobs subcategories (matching legacy Jobs categories)
- Configure specifications on each subcategory:
  - Employment Type (string select)
  - Education Level (string select)
  - Years of Experience (string select)
  - Company Perks (multiple string select slots, one per perk)
- Publish to production (behind feature flag in GraphQL)

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6. Frontend — Web (Next.js)

Requires frontend developer (not yet assigned)

Updates Required

6.1 SRP

- Add "Jobs" as a `marketType` filter option
- Render Jobs specSubCat filters when Jobs is selected
- Custom pay range filter UI (hourly/salary toggle + range)
- Grid vs list display rules (frontend-only logic)

6.2 Detail Page

- Render Jobs top-level fields: pay range, Apply button
- specSubCat fields render through existing specs UI

6.3 MyAccount

- Display Jobs listings
- Support actions: view, mark as sold, mark as sale pending, delete, renew, reactivate
- No inline editing — editing via PAL (out of scope)

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7. App (React Native) — 2-week estimate

Capacity not confirmed with App team

Updates Required

- Jobs tab: auto-select Jobs category, show subcategory list (same as Services)
- Remove Jobs from global search
- Detail Page: render pay range + Apply button for Jobs listings
- PAL: **scope TBD** — confirm whether hard-coded Jobs fields in app PAL are included

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Services NOT Changed in Phase 2

- `listing-applications` — Quick Apply is Phase 4
- `listing-feed-parser` / `listing-feed-subscriber` — Feed ingestion unchanged
- `listing-cron-boosts` / `listing-cron-stats` — No changes needed
- Saved Search services — Phase 3